

The ground-state measure is the time-slice marginal of the Euclidean measure, and that imports the local large-field estimate

The obstruction recorded in the identities note is removed by an identification rather than by an estimate. For the Kogut–Susskind Hamiltonian obtained as the $a_t \rightarrow 0$ limit of the anisotropic Wilson transfer matrix,

$$|\Omega(U)|^2 d\mu(U) = \lim_{T \rightarrow \infty} \text{marginal at time 0 of } \frac{e^{-S_E[\mathcal{U}]/\hbar} \prod d\mu}{Z_T},$$

so the ground-state measure of the Hamiltonian theory is the distribution of one time slice under the four-dimensional Euclidean Wilson measure. Every estimate proved for that measure therefore holds for $|\Omega|^2$ verbatim, including the local ones that the eigenvalue equation cannot reach, because the Euclidean weight $e^{-S_E/\hbar} = \prod_P e^{-\beta_P/\hbar}$ factorizes over four-dimensional plaquettes while the eigenvalue equation carries the extensive E_0 . The estimate the route needs is then available in its standard form: a configuration whose field strength averaged over a block of side ℓ reaches η/ℓ^2 costs Euclidean action

$$\frac{S_E}{\hbar} \simeq \frac{1}{4g^2} \int_{\text{block}} |F|^2 d^4x \simeq \frac{\eta^2}{4g^2},$$

independent of ℓ and of the lattice spacing, so the suppression is $e^{-c\eta^2/g(\ell)^2}$ at every scale, with the running coupling. This matches, with the same exponent structure, the free-field Gaussian tail $\exp[-8\pi^2\eta^2/g^2]$ computed in the typical-field note §3, which is a check on both. With the identification, the decimation step of the route controls its truncation error off a set of measure $e^{-c\eta^2/g^2}$, and what remains is the treatment of that set: the large-field problem in its usual form, now reached honestly rather than assumed. Constants explicit; the identification is standard and cited; nothing promoted.

1. The transfer matrix and the limit

On an anisotropic lattice with spatial spacing a and temporal spacing a_t , the Wilson action separates into temporal and spatial plaquettes,

$$\frac{S_E}{\hbar} = \frac{1}{g^2} \left[\frac{a}{a_t} \sum_{P \text{ temporal}} (N - \text{Re tr } U_P) + \frac{a_t}{a} \sum_{P \text{ spatial}} (N - \text{Re tr } U_P) \right] \cdot \frac{2}{1},$$

in the normalization of the obligations map with g dimensionless. The transfer matrix $\mathcal{T}$ of this action is self-adjoint and strictly positive (Lüscher, Commun. Math. Phys. 54 (1977) 283; Creutz, Phys. Rev. D 15 (1977) 1128; both metadata level), and as $a_t \rightarrow 0$

$$\mathcal{T} = \exp \left[-\frac{a_t}{\hbar} H_{\text{KS}} \right] (1 + O(a_t)), \quad H_{\text{KS}} = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p) \right],$$

the temporal plaquettes producing the electric term through the character expansion and the spatial ones passing directly to the magnetic term.

Proposition 1. Let Ω be the Perron vector of $\mathcal{T}$, which by T1 of the obligations map is the positive ground state of H_{KS} in the limit. Then for any bounded F of a single time slice,

$$\langle \Omega, F \Omega \rangle = \lim_{T \rightarrow \infty} \frac{\langle \chi, \mathcal{T}^T F \mathcal{T}^T \chi \rangle}{\langle \chi, \mathcal{T}^{2T} \chi \rangle} = \lim_{T \rightarrow \infty} \langle F \rangle_{S_E, [-T, T]},$$

for any χ with $\langle \chi, \Omega \rangle \neq 0$; that is, $|\Omega|^2 d\mu$ is the time-slice marginal of the Euclidean measure.

Proof. Spectral decomposition of $\mathcal{T}$ with the gap of T1 gives the first equality; the middle expression is by construction the Euclidean expectation of F inserted at time 0 in a slab of height $2T$ with boundary state χ , which is the second. $\square$

Nothing here is new; the point is what it licenses.

2. What transfers

Any statement of the form “the Euclidean measure assigns probability at most p to the set of configurations whose restriction to a time slice has property $\mathcal{A}$ ” is, by Proposition 1, a statement about $|\Omega|^2$. In particular the local statements do, and the reason the Hamiltonian formulation could not produce them itself is now visible as a feature of the representation rather than of the physics: the eigenvalue equation $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ carries the extensive e_0 and compares totals, while $e^{-S_E/\hbar}$ is a product over plaquettes and compares locally. Both describe the same measure.

This closes the gap left open in the identities note §5 and in the typical-field note §3, in the only way those notes left available.

3. The large-field estimate at a block scale

The action cost is scale-invariant. Let a configuration have field strength of magnitude η/ℓ^2 coherently over a four-dimensional block of side ℓ . Its Euclidean action is

$$\frac{S_E}{\hbar} = \frac{1}{4g^2} \int |F_{\mu\nu}^a|^2 d^4x \simeq \frac{1}{4g^2} \left(\frac{\eta}{\ell^2}\right)^2 \ell^4 = \frac{\eta^2}{4g^2},$$

independent of ℓ and of a . This is the reason large-field estimates take the same form at every scale, with only the coupling running: the suppression is

$$\mathbb{P}\left(|F|_{\text{block}} \geq \frac{\eta}{\ell^2}\right) \lesssim \exp\left[-\frac{c\eta^2}{g(\ell)^2}\right],$$

with c a pure number and $g(\ell)$ the coupling at the block scale.

Consistency check. The free-field computation of the typical-field note §3 gave, for the flowed magnetic field at radius $\sqrt{8t} = \ell$,

$$\mathbb{P}\left(|b_t| > \frac{\eta}{t}\right) \simeq \exp\left[-\frac{8\pi^2\eta^2}{g^2}\right],$$

the same η^2/g^2 in the exponent, obtained by an independent route from the Gaussian ground state. The two agree in structure and fix $c = 8\pi^2$ in the free limit.

Why the lattice-scale version is empty. At $\ell = a$ the plaquette angles are of order one, the action per plaquette is bounded by $4N/g^2$, and the Chebyshev bound $\mathbb{P}(S_R \geq s) \leq \exp[-s + 4N|R|/g^2]$ is vacuous, as noted in the typical-field note. The estimate has content only for the block-averaged field at $\ell \gg a$, where the individual plaquette angles are small, of order $a^2\eta/\ell^2$, and the cost accumulates over $(\ell/a)^4$ of them to the scale-invariant total above. The large-field condition is a statement about coherence across a block, not about any single plaquette.

Superseded in part. The competition below between $e^{-c\eta^2/g^2}$ and $e^{2\eta}$ holds at fixed truncation range; letting the range grow like η removes it, and the real obstruction in the large-field region is the sign of the error, see the operator-inequality note.

4. Where this leaves the route

With Proposition 1 and Section 3, the decimation step of the position note §6 has its error controlled outside a set of ground-state measure $e^{-c\eta^2/g(\ell)^2}$, which at weak coupling is small, and inside that set the flow-truncation bound of the Jacobian note degrades like $e^{2t\|G\|_\infty}$. The two exponents compete:

$$\text{gain } e^{-c\eta^2/g^2} \quad \text{against} \quad \text{loss } e^{2\eta},$$

since $t\|G\|_\infty \simeq \eta$ when the field reaches η/ℓ^2 at the flow radius. The gain wins for

$$\eta \gtrsim \frac{2g^2}{c},$$

so the large-field region can be defined as $\{|F|_{\text{block}} \geq \eta_*/\ell^2\}$ with $\eta_* = 2g^2/c$, on which the measure is at most $e^{-4g^2/c}$ and outside which the truncation is accurate. **At weak coupling** $\eta_* \rightarrow 0$, **so the good region is almost everything**; the residual set carries measure $e^{-4g^2/c}$, which tends to one as $g \rightarrow 0$, and there the estimate fails to be useful.

That last sentence is the honest statement of where the difficulty now sits: the competition is between an exponential gain in η^2/g^2 and an exponential loss in η , and at small g the crossover η_* is small, so the excluded set is defined by a weak condition and its measure bound $e^{-4g^2/c}$ is close to one. Improving it requires a sharper treatment of the truncation inside the large-field region, which is the large-field expansion of the constructive programme, or a truncation whose error grows more slowly than $e^{2t\|G\|_\infty}$.

5. Consequence for STATE

The identification of $|\Omega|^2$ with the Euclidean time-slice marginal is recorded, with the transfer-matrix references, and it imports the local estimates that the Hamiltonian eigenvalue equation could not produce. The large-field estimate itself has the scale-invariant form $e^{-c\eta^2/g^2}$, confirmed against the independent free-field computation. The route's decimation step now has a quantitative structure: good region with accurate truncation, bad region of measure $e^{-c\eta_*^2/g^2}$, and a crossover η_* fixed by the competition between the two exponentials. The next question is the first one that is genuinely about the large-field region rather than around it: can the flow-truncation error inside it be bounded by something weaker than $e^{2t\|G\|_\infty}$, for instance by using that the flow contracts the action monotonically even there?